INFORMATION TO KEEP WITH YOU

My complete treatment handbook

Docetaxel + pertuzumab + trastuzumab

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
21 days
Intervals differ by treatment phase

Treatment stages
2
Explained in the treatment reference

Regimen source
NCCP 00204
Version 11

intravenous (IV) pertuzumab, IV trastuzumab and docetaxel every 21 days. Docetaxel duration is reviewed separately; the two HER2 antibodies may continue after chemotherapy stops while beneficial and tolerated.

Treatment stages

01 **Current combination**
21-day cycle

02 **Antibody-only phase, if prescribed**
21-day cycle

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name _____

Hospital / oncology team _____

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Report new breathlessness, chest symptoms, severe diarrhoea, dehydration, jaundice, unusual bleeding or sudden illness promptly. Do not wait for the next planned test.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

What each medicine does

Control the cancer and symptoms, with repeated review of benefit and tolerance.

TAXANE CHEMOTHERAPY

Docetaxel



Disrupts the cell’s division machinery

Docetaxel stabilises tiny structures called microtubules. Cells need these to separate their chromosomes during division; disrupting the process can stop cancer cells multiplying. Healthy dividing cells are also affected.

Microtubules stabilised → Division disrupted → Cell growth affected

What to watch for

Take prescribed steroid premedication. Report fever, lasting numbness, swelling, sore nails or infusion symptoms.

HER2-TARGETED ANTIBODY

Pertuzumab



Blocks HER2 pairing with other receptors

Pertuzumab binds a different area of HER2 from trastuzumab. It limits receptor pairing, so the two antibodies act together at different points on the same growth pathway.

HER2 pairing → Pertuzumab blocks pairing → Growth signals reduced

What to watch for

Report diarrhoea early, especially with poor fluid intake. Keep cardiac checks and report infusion symptoms immediately.

HER2-TARGETED ANTIBODY

Trastuzumab



Targets HER2 growth signals

Trastuzumab binds to the HER2 protein on cells and helps reduce HER2-driven growth and recruit immune activity. HER2 testing identifies the cancers for which this approach is used.

HER2 on cell surface → Antibody binds → Growth signalling reduced

What to watch for

Keep echo appointments throughout HER2 treatment. Report breathlessness, swelling, palpitations or a reaction; chemotherapy side effects are tracked separately.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Why this treatment?

This combines docetaxel chemotherapy with two HER2-targeted antibodies for HER2-positive metastatic or unresectable locally recurrent breast cancer. The NCCP indication applies before chemotherapy or HER2 treatment for metastatic disease. The team considers earlier treatment, general health, heart function, symptoms and alternatives. Chemotherapy and later antibody-only maintenance are distinct phases.

What could change the choice?

Ask which cancer features, earlier treatments and health factors led to this recommendation, which alternatives were considered and when the decision will be reviewed.

Your treatment and the next step

Schedule

intravenous (IV) pertuzumab, IV trastuzumab and docetaxel every 21 days. Docetaxel duration is reviewed separately; the two HER2 antibodies may continue after chemotherapy stops while beneficial and tolerated.

After this phase

Continue while treatment is helping and tolerated. If a component stops, confirm the exact medicines, tests and supportive prescription for the new phase. Maintenance remains active treatment.

CHAPTER 02

Your schedule, tests and appointments

IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

The phases and how they connect

Treatment stages

01 **Current combination**
21-day cycle

02 **Antibody-only phase, if prescribed**
21-day cycle

Your team confirms the active stage and dates. The full treatment guide explains each step.

This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

21-DAY CYCLE
Current combination

21-day cycle

intravenous (IV) pertuzumab, IV trastuzumab and docetaxel every 21 days. Docetaxel duration is reviewed separately; the two HER2 antibodies may continue after chemotherapy stops while beneficial and tolerated.

21-DAY CYCLE
Antibody-only phase, if prescribed

21-day cycle

Continue while treatment is helping and tolerated. If a component stops, confirm the exact medicines, tests and supportive prescription for the new phase. Maintenance remains active treatment.

Relative sequence, not equal lengths or booked dates. Your team confirms actual appointments and the active medicines.

BASELINE → TREATMENT → FOLLOW-UP

Tests and milestones through treatment

Columns show sequence, not equal durations or appointments. A test is only complete when performed and reviewed. Echo timing and cancer-imaging timing are different. Symptoms can bring any check forward.

Check	Before starting	First cycles	Around week 12	Continuing / next phase	Longer-term plan
Blood tests	Baseline full blood count (FBC), renal and liver profile	Before each cycle; additional dose checks as prescribed	Continue the required pre-dose checks	Use the selected active regimen's blood-test schedule	Investigate symptoms / persistent abnormalities

Check	Before starting	First cycles	Around week 12	Continuing / next phase	Longer-term plan
Echo / cardiac assessment	Baseline cardiac assessment	Earlier for symptoms / concern	Echo / multigated acquisition (heart pumping) scan (MUGA) due around 12 weeks	Continue every 12 weeks; keep the original clock	Agree follow-up for the active medicine and cardiac risk
Cancer imaging / markers	Baseline disease assessment	Clinical response; imaging if concern	Response review as planned; not tied to echo	Reassess disease and treatment plan	Imaging generally every 2–4 months; individual plan
Surgery / radiotherapy	Agree indication and sequence	No automatic concurrent radiation	Coordinate with both teams	Local treatment as indicated	Review recovery and late effects
Endocrine / bone health	Check hormone receptors / fracture risk	Plan endocrine therapy if indicated	dual-energy X-ray absorptiometry (bone density) (DEXA) when relevant to bone-risk plan	Endocrine timing by the treating team	Individual DEXA / bone follow-up

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00204 v11 · official regimen · ESMO 2024 early breast cancer guideline · treatment and follow-up · ESC 2022 cardio-oncology guideline · cardiac surveillance and risk · NICE NG101 · bone health and follow-up · ESMO metastatic breast cancer guideline · 2026

THE WHOLE TREATMENT JOURNEY

Docetaxel + pertuzumab + trastuzumab: the whole pathway

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	First cycles	Around week 12	Continuing / next phase	Longer-term plan
Cancer medicines NCCP	Confirm diagnosis and exact regimen	See the dose-day map: use the current regimen and active phase	Echo due; treatment review	Review active medicines and benefit / tolerance	Review benefit / tolerance and next plan
Surgery / radiotherapy METASTATIC	Agree indication and sequence	No automatic concurrent radiation	Coordinate with both teams	Local treatment as indicated	Review recovery and late effects
Blood tests NCCP	Baseline FBC, renal and liver profile	Before each cycle; additional dose checks as prescribed	Continue the required pre-dose checks	Use the selected active regimen's blood-test schedule	Investigate symptoms / persistent abnormalities
Echo / cardiac checks NCCP · ESC	Baseline cardiac assessment	Earlier for symptoms / concern	Echo / MUGA due around 12 weeks	Continue every 12 weeks; keep the original clock	Agree follow-up for the active medicine and cardiac risk
Scans / tumour markers METASTATIC · MARKERS	Baseline disease assessment	Clinical response; imaging if concern	Response review as planned; not tied to echo	Reassess disease and treatment plan	Imaging generally every 2–4 months; individual plan
Endocrine / DEXA BONE	Check hormone receptors / fracture risk	Plan endocrine therapy if indicated	DEXA when relevant to bone-risk plan	Endocrine timing by the treating team	Individual DEXA / bone follow-up

What each check is for

1

Blood tests

FBC, kidney and liver profile at baseline and before every 21-day cycle. The team reviews the necessary results before a planned dose.

Check safety before the next dose.

Ask / act: Agree the booking owner, result reviewer and action if overdue.

2

Heart tests

Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock. After the first year, ESC guidance allows six-monthly echo in selected asymptomatic low- or moderate-risk metastatic patients with normal assessments, only under an agreed team plan. Entering maintenance alone does not justify reducing checks.

Monitor pumping function and relevant rhythm risk.

Ask / **act:** Agree the booking owner, result reviewer and action if overdue.

3



Cancer imaging

Record baseline disease sites. Response imaging is generally reviewed every 2–4 months under international guidance, adapted to symptoms, disease pace and treatment. The team selects the scan; this is not an automatic computed tomography (CT) chest/abdomen/pelvis order for every patient. Tumour markers may support assessment in selected cases, but do not alone decide progression or a treatment change.

Assess response or follow-up needs.

Ask / **act:** Agree the booking owner, result reviewer and action if overdue.

4

Bone health

If the cancer is also hormone-receptor positive, endocrine treatment may form part of the wider plan. Bone-density testing is linked to aromatase inhibitors, ovarian suppression and fracture risk; it is not automatically required because a HER2 medicine is prescribed. Agree baseline DEXA where indicated and individual follow-up.

Assess fracture risk where relevant.

Ask / **act:** Agree the booking owner, result reviewer and action if overdue.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Your practical treatment plan



STEP 1

Before the first visit

Confirm blood tests, heart assessment, active medicines, route and the consent discussion.

STEP 2

Plan venous access

A cannula, peripherally inserted central catheter (PICC) or implanted port may be used. If a port is chosen, a trained team inserts it in a separate procedure with local anaesthetic and sometimes sedation; the team explains fasting, blood-thinner and wound instructions.

STEP 3

Infusion day

Staff check the line, give prescribed premedicines and treatment, and observe for reactions. Tell them immediately about burning, pain, rash or breathlessness.

STEP 4

Home and next review

This regimen does not routinely need a take-home infusion pump. Staff disconnect the infusion; a port is de-accessed when appropriate. Keep line-care and next-visit instructions.

Port insertion and removal

The line team explains placement, dressing care and when normal activities can resume. A port is accessed with a special needle during treatment. Removal of the implanted device is a separate planned procedure when no longer needed; do not remove or adjust it yourself.

PICC care and urgent line symptoms

A PICC usually needs regular dressing and flushing appointments even between treatments. Fever, redness, swelling, pain, leakage or damage requires prompt advice. Your hospital supplies the exact care instructions.

Visit time and transport

First visits can take longer because of loading doses, premedicines and observation. Ask the unit about expected time, transport and any escort needed after sedation or line insertion.

Phase-specific visit pattern

Infusion and observation times differ by drug and first versus later doses; the unit confirms the full visit duration, including blood review and preparation.

When scans, heart checks and bloods happen

Agree who books each test, who reviews the result and what happens if a check is overdue.

Blood tests

FBC, kidney and liver profile at baseline and before every 21-day cycle. The team reviews the necessary results before a planned dose.



Heart ultrasound and electrocardiogram (ECG)

Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock. After the first year, ESC guidance allows six-monthly echo in selected asymptomatic low- or moderate-risk metastatic patients with normal assessments, only under an agreed team plan. Entering maintenance alone does not justify reducing checks.

Scans and tumour markers

Record baseline disease sites. Response imaging is generally reviewed every 2–4 months under international guidance, adapted to symptoms, disease pace and treatment. The team selects the scan; this is not an automatic CT chest/abdomen/pelvis order for every patient. Tumour markers may support assessment in selected cases, but do not alone decide progression or a treatment change.

Bone health and endocrine treatment

If the cancer is also hormone-receptor positive, endocrine treatment may form part of the wider plan. Bone-density testing is linked to aromatase inhibitors, ovarian suppression and fracture risk; it is not automatically required because a HER2 medicine is prescribed. Agree baseline DEXA where indicated and individual follow-up.

How surgery and radiotherapy fit

Surgery or radiotherapy may be used for particular symptoms or disease sites. There is no routine surgery-then-radiation sequence for this advanced-cancer regimen. The oncology and radiation teams agree any treatment pause or overlap.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

Chemotherapy with HER2 treatment

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Chemotherapy with HER2 treatment

Treatment or check	Timing in this view
HER2 medicine · Day 1	 **Day 1** IV
Chemotherapy · Day 1	 **Day 1** IV
Administration reaction	 **Days 1–21** Report during or after any visit
Nausea / appetite	 **Days 1–5** May occur early or later
Blood counts / infection	 **Days 1–21** Risk varies; fever needs prompt advice
Fatigue / mouth changes	 **Days 1–21** Variable timing and severity
Nerves / skin / nails	 **Days 1–21** Can build across doses
Heart / breathing symptoms	 **Days 1–21** Report on any day

Planned treatment: IV pertuzumab, IV trastuzumab and docetaxel every 21 days. Docetaxel duration is reviewed separately; the two HER2 antibodies may continue after chemotherapy stops while beneficial and tolerated.

Heart checks: Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock. After the first year, ESC guidance allows six-monthly echo in selected asymptomatic low- or moderate-risk metastatic patients with normal assessments, only under an agreed team plan. Entering maintenance alone does not justify reducing checks.

Next phase: Continue while treatment is helping and tolerated. If a component stops, confirm the exact medicines, tests and supportive prescription for the new phase. Maintenance remains active treatment.

Why these bands stay broad: Treatment markers show the planned schedule. Symptom bands are broad possibilities, not measured risk curves; serious symptoms can occur at any time. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00204 v11 · official regimen ↗

There are no guaranteed "safe days". Fever, shivers, sudden illness, severe breathing changes or chest pain need prompt assessment on any day.

HER2-only continuation · source 21-day schedule

Day 1 → Day 21: a general pattern

HER2-only continuation · source 21-day schedule

Treatment or check	Timing in this view
Treatment · Day 1	 **Day 1** IV
Administration reaction	 **Days 1–21** Report during or after any visit
Heart / breathing symptoms	 **Days 1–21** Report on any day

Planned treatment: IV pertuzumab, IV trastuzumab and docetaxel every 21 days. Docetaxel duration is reviewed separately; the two HER2 antibodies may continue after chemotherapy stops while beneficial and tolerated.

Heart checks: Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock. After the first year, ESC guidance allows six-monthly echo in selected asymptomatic low- or moderate-risk metastatic patients with normal assessments, only under an agreed team plan. Entering maintenance alone does not justify reducing checks.

Next phase: Continue while treatment is helping and tolerated. If a component stops, confirm the exact medicines, tests and supportive prescription for the new phase. Maintenance remains active treatment.

> **Why these bands stay broad:** Treatment markers show the planned schedule. Symptom bands are broad possibilities, not measured risk curves; serious symptoms can occur at any time. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00204 v11 · official regimen ↗

Between-treatment notes

Day 1–21: The current treatment cycle

IV pertuzumab, IV trastuzumab and docetaxel every 21 days. Docetaxel duration is reviewed separately; the two HER2 antibodies may continue after chemotherapy stops while beneficial and tolerated. Your team confirms treatment dates and checks. Serious symptoms need advice on any day.

Support around the medicines

1 

The correct supportive prescription

Follow the script for the active medicine, route and treatment day. Do not carry chemotherapy anti-sickness medicines automatically into an antibody-only phase.

2

Keep heart checks on the plan

An echo every 12 weeks is separate from blood tests or cancer imaging. Ask who books and reviews it.

3

Plan for symptoms

Keep your agreed contact numbers and instructions. New symptoms can need assessment before a routine appointment.

Questions worth asking

Who arranges my tests?

Ask the team who books each test, who checks the result and who to contact if an appointment is overdue. This map does not book appointments.

What if I become unwell between visits?

Use the urgent contact plan for sudden illness, fever, chest symptoms, severe diarrhoea or poor fluid intake. Do not wait for a scheduled blood test or scan.

> **Try explaining the plan back:** Which medicines are active now, which days do you receive or take them, and who is booking and reviewing your next tests?

Docetaxel

- Low blood counts, infection and sepsis
- Hair loss, mouth soreness and fatigue
- Numbness / tingling; sometimes lasting
- Fluid retention and nail changes
- Allergic infusion reaction

Pertuzumab

- Diarrhoea and dehydration
- Infusion / allergic reaction
- Heart pumping-function reduction
- Rash and fatigue

Trastuzumab

- Heart pumping-function reduction / heart failure
- Infusion or allergic reaction
- Diarrhoea and fatigue
- Rare serious lung effects

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

Docetaxel + HER2 antibodies

Active medicines: Docetaxel + Pertuzumab + Trastuzumab

Emetogenic risk: Low

Medicine	Dose / route	When
Ondansetron	8 mg orally	On treatment day Review QT, constipation and hepatic dosing; reconcile any steroid already prescribed

Medicine	Dose / route	When
Metoclopramide	10 mg orally up to three times daily as needed	Subsequent days: duration to be confirmed Assess contraindications, renal function and interactions

Additional phase-specific support

- Docetaxel: dexamethasone 8 mg orally twice daily for 3 days starting the day before treatment; minimum 3 doses pre-treatment unless contraindicated. If oral doses were missed, any 20 mg intravenous (IV) replacement is a consultant decision.
- Reconcile docetaxel premedication with the antiemetic steroid course and interactions; issue one coherent steroid prescription, not duplicate courses.
- Consider primary granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) using the source and individual risk; product and timing follow the prescribed plan.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.
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- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

Antibody-only continuation

Active medicines: Pertuzumab + Trastuzumab

Emetogenic risk: Minimal

No routine prophylactic antiemetic order is populated for this phase. Agree an individual symptom-treatment plan where needed.

Additional phase-specific support

- No routine taxane dexamethasone, chemotherapy granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) or cytotoxic antiemetic course carries forward.
- Review previous reactions and individual rescue needs; keep cardiac checks on the existing clock.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

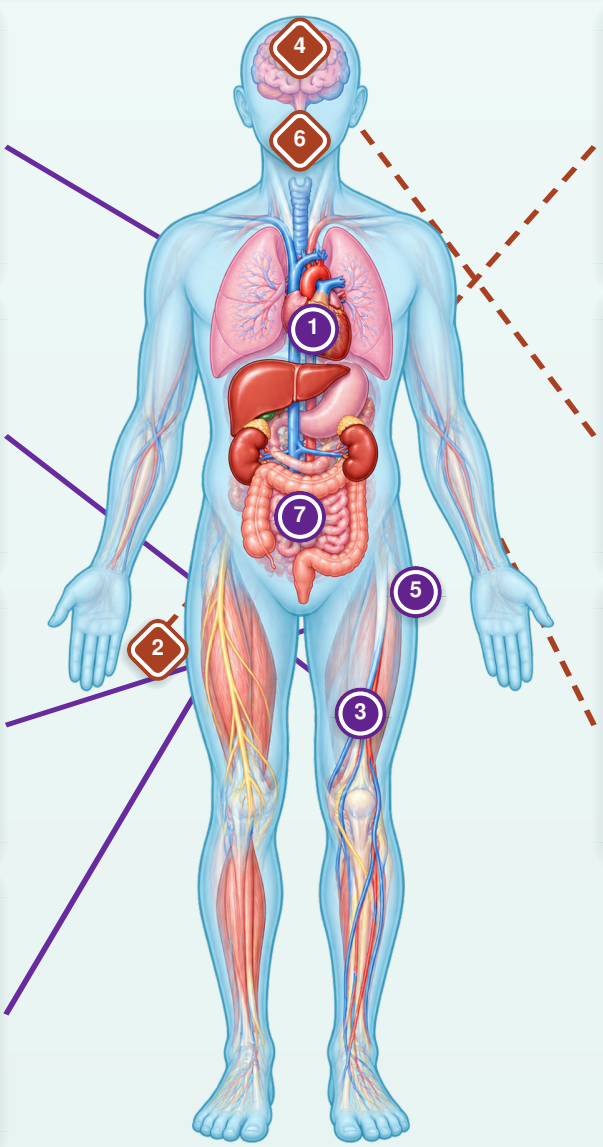
Docetaxel + pertuzumab + trastuzumab

intravenous (IV) pertuzumab, IV trastuzumab and docetaxel every 21 days. Docetaxel duration is reviewed separately; the two HER2 antibodies may continue after chemotherapy stops while beneficial and tolerated.

Source reconciled; independent clinical, pharmacy and patient/PPI review pending. Selected effects; not a complete medicine leaflet.

A picture of side effects

Numbered symptom boxes connect to the anatomical illustration. The medicine responsible may be uncertain; report the symptom.



1 Heart

Trastuzumab / pertuzumab

Keep echo appointments; report breathlessness or swelling.

2 Hands / feet

Docetaxel

Report numbness or tingling before the next dose.

3 Blood / infection

Chemotherapy phase

Fever, shivers or suddenly unwell: call immediately.

4 Hair / scalp

Chemotherapy phase

Hair loss is common. Ask about scalp cooling before treatment where suitable.

5 Fluid / skin / nails

Docetaxel

Report swelling, painful skin or nail changes.

6 Mouth

Chemotherapy phase

Painful ulcers or difficulty drinking need advice.

7 Bowel

Pertuzumab / chemotherapy

Report diarrhoea promptly, especially with poor intake.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

The map shows selected effects of the medicines named here. Report persistent effects from earlier treatment too.

> Fever, shivers or suddenly unwell: contact oncology immediately. Severe chest pain, breathing difficulty, collapse or seizure: 112 / 999.

CHAPTER 04

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **National Cancer Control Programme (NCCP) 00204 v11 · official regimen**
   https://healthservice.hse.ie/documents/6545/204_v11_PertuzumabTrasDoc.pdf
2. **ESMO 2024 early breast cancer guideline · treatment and follow-up**
   https://www.annalsofoncology.org/article/S0923-7534(23)05104-9/fulltext
3. **ESC 2022 cardio-oncology guideline · cardiac surveillance and risk**
   https://academic.oup.com/eurheartj/article/43/41/4229/6673995
4. **NICE NG101 · bone health and follow-up**
   https://www.nice.org.uk/guidance/ng101/chapter/Recommendations
5. **ESMO metastatic breast cancer guideline · 2026**
   https://www.annalsofoncology.org/article/S0923-7534(26)00902-6/fulltext
6. **ASCO · biomarkers in metastatic breast cancer**
   https://ascopubs.org/doi/10.1200/JCO.2015.61.1459
7. **EMA Herceptin product information**
   https://www.ema.europa.eu/en/documents/product-information/herceptin-epar-product-information_en.pdf
8. **eviQ · implanted port patient information**
   https://www.eviq.org.au/p/507
9. **eviQ · peripherally inserted central catheter (PICC) patient information**
   https://www.eviq.org.au/p/543
10. **HSE · chemotherapy side effects and urgent advice**
    https://www2.hse.ie/conditions/chemotherapy/side-effects/
11. **NOGG · section 7, aromatase inhibitors and fracture risk**
    https://www.nogg.org.uk/full-guideline/section-7-strategies-management-osteoporosis-and-fracture-risk
12. **NCCP NCIS antiemetics v8 (2025), Table 1**
    https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf
13. **EMA Perjeta product information**
    https://www.ema.europa.eu/en/documents/product-information/perjeta-epar-product-information_en.pdf